

## Topic 2 – Membranes, Proteins, DNA and Gene Expression

Candidates will be assessed on their ability to:

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| <b>2.1</b>                                                                                                                                                               | <ul style="list-style-type: none"> <li>(i) know the properties of gas exchange surfaces in living organisms (large surface area to volume ratio, thickness of surface and difference in concentration)</li> <li>(ii) understand how the rate of diffusion is dependent on these properties and can be calculated using Fick's Law of Diffusion</li> <li>(iii) understand how the structure of the mammalian lung is adapted for rapid gaseous exchange</li> </ul>                                                                                                                                                                                                                                                                                                            |
| <b>2.2</b>                                                                                                                                                               | <ul style="list-style-type: none"> <li>(i) know the structure and properties of cell membranes</li> <li>(ii) understand how models such as the fluid mosaic model of membrane structure are interpretations of data used to develop scientific explanations of the structure and properties of cell membranes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>2.3</b>                                                                                                                                                               | <p><b>CORE PRACTICAL 3</b></p> <p><b>Investigate membrane properties including the effect of alcohol and temperature on membrane permeability.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>2.4</b>                                                                                                                                                               | understand what is meant by osmosis in terms of the movement of free water molecules through a partially permeable membrane, down a water potential gradient                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>RECOMMENDED ADDITIONAL PRACTICAL</b></p> <p><b>Investigate tissue water potentials using plant tissue and graded concentrations of a solute.</b></p>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>2.5</b>                                                                                                                                                               | <ul style="list-style-type: none"> <li>(i) understand what is meant by passive transport (diffusion, facilitated diffusion), active transport (including the role of ATP as an immediate source of energy), endocytosis and exocytosis</li> <li>(ii) understand the involvement of carrier and channel proteins in membrane transport</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>2.6</b>                                                                                                                                                               | <ul style="list-style-type: none"> <li>(i) know the basic structure of an amino acid<br/><i>Structures of specific amino acids are not required.</i></li> <li>(ii) understand the formation of polypeptides and proteins (amino acid monomers linked by condensation reactions to form peptide bonds)</li> <li>(iii) understand the significance of a protein's primary structure in determining its secondary structure, three-dimensional structure and properties (globular and fibrous proteins and the types of bonds involved in its three-dimensional structure)</li> <li>(iv) know the molecular structure of a globular protein and a fibrous protein and understand how their structures relate to their functions (including haemoglobin and collagen)</li> </ul> |
| <p><b>RECOMMENDED ADDITIONAL PRACTICAL</b></p> <p><b>Use a semi-quantitative method to estimate protein concentration using biuret reagent and colour standards.</b></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>2.7</b>                                                                                                                                                               | <ul style="list-style-type: none"> <li>(i) understand the mechanism of action and the specificity of enzymes in terms of their three-dimensional structure</li> <li>(ii) understand that enzymes are biological catalysts that reduce activation energy</li> <li>(iii) know that there are intracellular enzymes catalysing reactions inside cells and extracellular enzymes catalysing reactions outside cells</li> </ul>                                                                                                                                                                                                                                                                                                                                                   |

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| <b>2.8</b>  | <b>CORE PRACTICAL 4</b><br><b>Investigate the effect of temperature, pH, enzyme concentration and substrate concentration on the initial rate of enzyme-catalysed reactions.</b>                                                                                                                                                                                                                                                                                             |
| <b>2.9</b>  | (i) know the basic structure of mononucleotides (deoxyribose or ribose linked to a phosphate and a base, including thymine, uracil, adenine, cytosine or guanine) and the structures of DNA and RNA (polynucleotides composed of mononucleotides linked by condensation reactions to form phosphodiester bonds)<br>(ii) know how complementary base pairing and the hydrogen bonding between two complementary strands are involved in the formation of the DNA double helix |
| <b>2.10</b> | (i) understand the process of DNA replication, including the role of DNA polymerase<br>(ii) understand how Meselson and Stahl's classic experiment provided new data that supported the accepted theory of replication of DNA and refuted competing theories                                                                                                                                                                                                                 |
| <b>2.11</b> | understand the nature of the genetic code (triplet code, non-overlapping and degenerate)                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>2.12</b> | know that a gene is a sequence of bases on a DNA molecule that codes for a sequence of amino acids in a polypeptide chain                                                                                                                                                                                                                                                                                                                                                    |
| <b>2.13</b> | (i) understand the process of protein synthesis (transcription and translation), including the role of RNA polymerase, translation, messenger RNA, transfer RNA, ribosomes and the role of start and stop codons<br>(ii) understand the roles of the DNA template (antisense) strand in transcription, codons on messenger RNA and anticodons on transfer RNA                                                                                                                |
| <b>2.14</b> | (i) understand how errors in DNA replication can give rise to mutations (substitution, insertion and deletion of bases)<br>(ii) know that some mutations will give rise to cancer or genetic disorders, but that many mutations will have no observable effect                                                                                                                                                                                                               |
| <b>2.15</b> | (i) understand what is meant by the terms <i>gene</i> , <i>allele</i> , <i>genotype</i> , <i>phenotype</i> , <i>recessive</i> , <i>dominant</i> , <i>codominance</i> , <i>homozygote</i> and <i>heterozygote</i><br>(ii) understand patterns of inheritance, including the interpretation of genetic pedigree diagrams, in the context of monohybrid inheritance<br>(iii) understand sex linkage on the X chromosome, including red-green colour blindness in humans         |
| <b>2.16</b> | understand how the expression of a gene mutation in people with cystic fibrosis impairs the functioning of the gaseous exchange, digestive and reproductive systems                                                                                                                                                                                                                                                                                                          |
| <b>2.17</b> | (i) understand the uses of genetic screening, including the identification of carriers, pre-implantation genetic diagnosis (PGD) and prenatal testing, including amniocentesis and chorionic villus sampling<br>(ii) understand the implications of prenatal genetic screening                                                                                                                                                                                               |
| <b>2.18</b> | be able to identify and discuss the ethical and social issues relating to genetic screening from a range of ethical viewpoints, including religious, moral and social implications                                                                                                                                                                                                                                                                                           |

# Unit 2: Cells, Development, Biodiversity and Conservation

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## IAS compulsory unit

## Externally assessed

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### Unit description

#### Introduction

This unit starts with the cell as the basic unit of all living organisms, leading to cell division, formation of gametes, fertilisation and the continuity of life. The roles of stem cells, gene expression, and the influence of the environment and epigenetics on phenotypes are also included. Cell development leads to an understanding of the structure and functions of plant cells, and how plants may be exploited by humans for fibres and as sources of drugs. This unit also considers the diversity of life and how biodiversity can be measured. The unit ends with an account of reasons for changes in populations over time, and the methods used by zoos and seed banks for the conservation of endangered species and their genetic diversity.

#### Practical skills

In order to develop their practical skills, students should be encouraged to carry out a range of core practical experiments related to this topic. There are also opportunities to carry out additional practical work, such as investigating factors affecting the growth of pollen tubes.

#### Mathematical skills

There are opportunities for the development of mathematical skills in this unit, including using ratios, percentages and fractions, substituting values into algebraic equations, calculation of magnification, understanding the terms *mean*, *mode* and *median*, constructing and interpreting frequency tables, bar charts and histograms. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

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## Assessment information

- First assessment: June 2019.
  - The assessment is 1 hour and 30 minutes.
  - The assessment is out of 80 marks.
  - Candidates must answer all questions.
  - The paper may include multiple-choice, short open, open-response, calculations and extended-writing questions.
  - The paper will include a minimum of 8 marks that target mathematics at Level 2 or above.
  - Calculators may be used in the examination. Please see *Appendix 8: Use of calculators*.
  - Candidates will be expected to apply their knowledge and understanding to familiar and unfamiliar contexts.
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